Project Proposal

You will need to submit a pdf document describing what you want to do. The document should contain the following sections:

- Title with project name
- Project summary/description (up to 100 words)
- Data sources table. Fill in the following table:

Title: Colon Cancer Survival Prediction through Machine Learning

Colon cancer was once considered a disease mainly affecting middle-aged or older adults, but recent studies have shown a rise in colon cancer among younger people. To address this change, building a machine learning (ML) prediction model can be helpful to estimate survival rates across different age groups. The model would use datasets containing patient age, global incidence colon cancer cases, cancer stage, risk factors, gender, and survival outcomes. After cleaning and organizing the data, these factors would train ML models such as logistic regression or decision trees to predict survival. The models would then be evaluated to assess how accurately they predict survival rates across age groups, genders, and countries. By comparing results between younger and older patients, the model could uncover patterns in survival outcomes.

Data Source #	Name/ Short Description	Source URL	Type: - API - Web page - File	List of Fields	Format: - json - xml - csv - sql - other	Have tried to access/collect data with python? yes/no	Estimated data size, number of data points you plan to use
1	NIH Colorectal Cancer Clinical Trials	https://clinicaltrialsapi.cancer.gov/api/v2	API	Healthcare	json	No (Not Yet)	300 or more
2	Colorectal Cancer Global Dataset	https://www.kaggle.com/datasets/ankushpanday2/colorectal-cancer-global-dataset-and-predictions/discussion	File	Healthcare	CSV	No (Not Yet)	300 or more

3	Colorectal Cancer Dietary and Lifestyle Dataset – colon cancer risk factors	https://www.kaggle.com/datasets/ziya07/colorectal-cancer-dietary-and-lifestyle-dataset	File	Healthcare	CSV	No (Not Yet)	300 or more
4	CDC Wonder	https://wonder.cdc.gov/wonder/help/wonder-api.html	Web	Healthcare	XML	No (Not Yet)	300 or more